

$$F_B = \frac{1}{2a} (q_a \cdot 3a + q_a \cdot \frac{a}{2})$$

$$= \frac{7}{4} q_a$$

$$(b) F = F_A + F_B + F_D$$

$$F \cdot a - F_B \cdot 2a - F_D \cdot 3a = 0$$

取 AC、CD、 $\downarrow q_a$

AC 是二力杆

C 点不会有水平力 $\therefore F_A = 0$

$$\sum M_B = q_a \cdot a + F_D \cdot a - q_a \cdot 2 = 0$$

$$\therefore F_D = 0 \quad F_B = \frac{7}{4} q_a$$

(a)

4-4 试作图4-2-6所示具有中间铰的梁的剪力图和弯矩图。

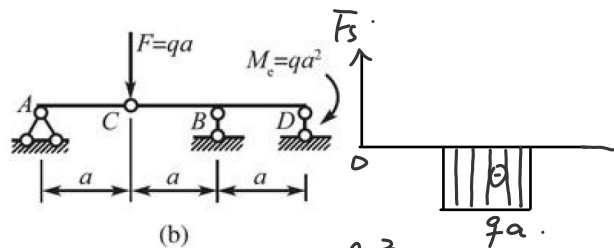
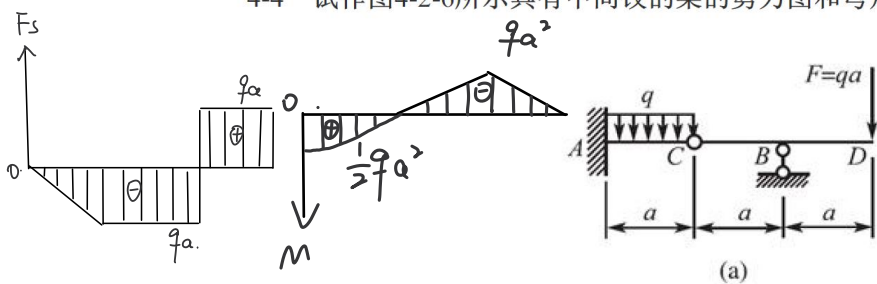


图4-2-6

提示：中间铰表明此处合力矩为0

4-8 试用叠加法作图4-2-14所示各梁的弯矩图。

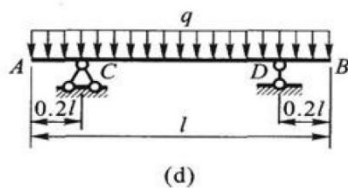
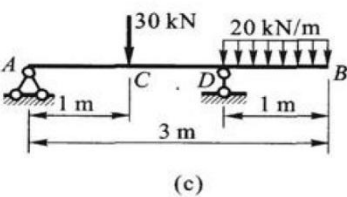
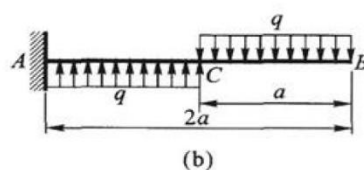
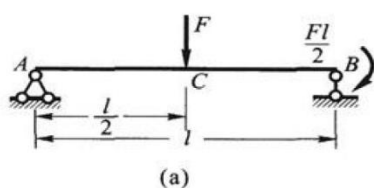
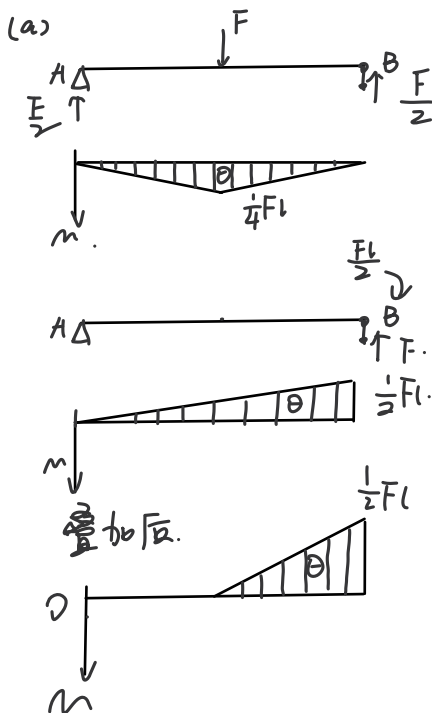
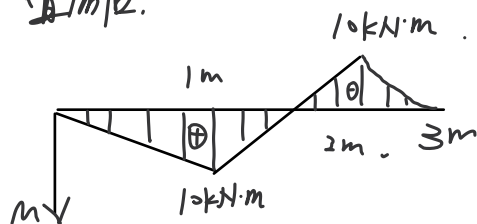
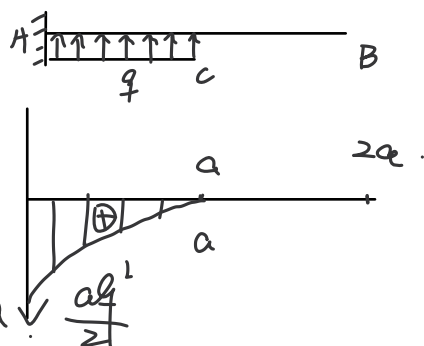
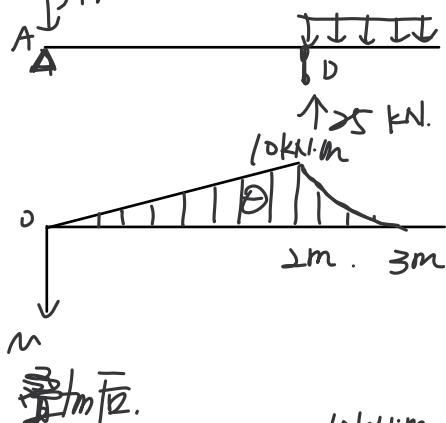
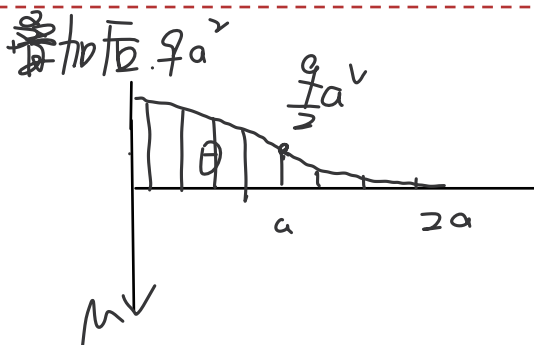
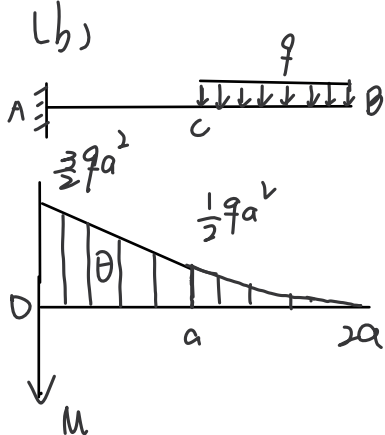
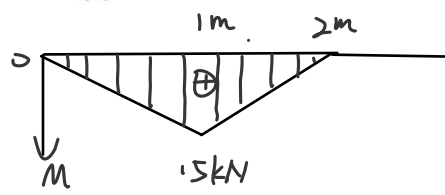
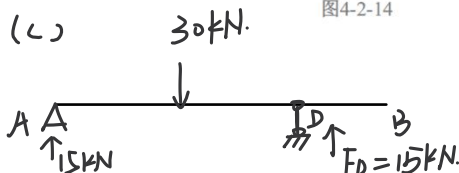


图4-2-14



(d)

